Pranav Nivarthi

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Professional Experience

Citigroup Technology May 2025 – Present
Generative AI Specialist | Contract via Xoriant *Tampa, FL*

- Led development of a reusable agentic AI platform serving **300+ daily active users** and **2,000+ total users** for enterprise search, analysis, presentations, and scheduled workflows.
- Cut presentation creation from **4+ hours to about 5 minutes** with containerized Node.js services supporting centrally managed and user-uploaded templates.
- Optimized LLM inference costs with a trained question-complexity router (**sub-50 ms observed latency**) and model escalation at configured response-time or LLM-call thresholds to support answer quality.
- Reduced observed session error rates to **near zero** and increased supported queries per second **about 10x without horizontal scaling** using a Couchbase event bus and Go executors.
- Implemented OpenTelemetry and Grafana monitoring, migrated LLM tracing from Langfuse to Google Cloud Observability, and built token usage, cost, and error analytics dashboards for business stakeholders.
- Improved factual correctness from **72% to 89%** using production data regularly annotated and verified by subject matter experts; scored completeness, entity search, and tool selection for regression checks and targeted improvements.
- Improved entity resolution accuracy from **80% to 97%** and answer consistency by **about 52%**, reducing average query response from **about 2 minutes to 40 seconds** through ongoing Neo4j GraphRAG semantic-layer work.
- Reduced agent time to production from **nearly a year to about 2–3 weeks** with shared MCP tools and a governance layer serving **8 downstream data platforms**, covering LDAP, centralized entitlements, audit trails, and data lineage.
- Strengthened trust and agent adoption among business users, technology teams, and SMEs through ontology-driven ingestion and custom deterministic file parsers for accurate, repeatable, and explainable extraction.

Citigroup Technology Nov 2023 – Apr 2025
Data Scientist | Contract via Xoriant *Pune, India*

- Supported financial anomaly detection by extracting **1,000+ financial metrics across 100+ pages** with **over 98% accuracy**, using document intelligence pipelines and Microsoft Table Transformer for complex tables.
- Enabled private enterprise knowledge access through RAG applications using locally hosted open-source LLMs and embeddings, model evaluation, semantic retrieval, and GraphRAG.
- Supported recovery from inference-job failures through PostgreSQL-backed asynchronous processing, retries, and state tracking for on-premises model serving.

ElasticRun Dec 2022 – Nov 2023
Associate Engineer – Data Science *Pune, India*

- Increased average invoice line items by **10%** with TIFUKNN/RACF recommendations influencing **55% of revenue**.
- Improved recommendation recall by **50%** through hyperparameter tuning, model feedback, and data reduction, supported by metric pipelines and Metabase dashboards.
- Increased telecaller conversion by **20%** with probabilistic contact-time recommendations and cut average call duration by **30%** with customer-profile dashboards.
- Reduced Databricks SQL compute costs and storage redundancy with Apache Pinot ETL for recommendation data.

AppViewX Feb 2021 – Dec 2022
QA Engineer *Remote*

- Cut regression effort by **57%** using Python Robot Framework to automate SSL/TLS lifecycle tests.

Technical Skills

Agents & LLMs: LangGraph, LangChain, Google ADK, Microsoft Agent Framework, MCP, tool calling, model routing
Retrieval: RAG, GraphRAG, embeddings, semantic search, knowledge graphs, ontology-driven ingestion
Evaluation & LLMOps: LLM evaluation, regression testing, model escalation, latency/token/cost optimization
Observability: OpenTelemetry, Grafana, Langfuse, Google Cloud Observability, distributed tracing, error analytics
Backend & Data: Python, Go, SQL, Node.js, Django, REST APIs, PostgreSQL, Neo4j, Couchbase, Databricks, Apache Pinot
Machine Learning: PyTorch, TensorFlow, Hugging Face, SVM, recommendation systems, anomaly detection, Pandas
Cloud & Governance: Docker, Kubernetes, Helm, AWS, Azure, Azure OpenAI, LDAP, entitlements, audit trails, data lineage

Education

Amrita School of Engineering, Coimbatore, India 2017 – 2021
B.Tech. in Electronics and Communication Engineering *GPA: 9.15/10 | Honors | University Rank 7*